

Axomeme 2.0: Deep Axial Transformer for Ultra-Fast Selection Inference

Architecture & Mathematical Specification

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Abstract

Axomeme 2.0 is a specialized axial attention transformer neural network designed for rapid, site-level evolutionary selection inference on multiple sequence alignments (MSAs) of protein-coding sequences. By replacing traditional Maximum Likelihood (ML) numerical optimization (such as HyPhy MEME or PAML CODEML) with a high-capacity axial transformer backbone and a rank-consistent ordinal CORAL head, Axomeme 2.0 infers Likelihood Ratio Test (LRT) selection statistics and evolutionary parameters in under **10 milliseconds per alignment** (a **4,000× speedup** over classical ML). This specification details the mathematical formulations, network components, attention mechanisms, and decoding algorithms.

1 System Overview

Given an input alignment matrix $\mathbf{X} \in \mathcal{C}^{N \times L}$ of N species (taxa) and L codon sites, Axomeme 2.0 maps sequence variation and phylogenetic distances directly to continuous site-level selection statistics $\hat{y}_{\text{LRT}} \in \mathbb{R}_{\geq 0}$ and selection classifications (Tier 1, Tier 2, or Neutral).

$$f_{\Theta}(\mathbf{X}, \mathbf{D}) \longrightarrow \left(\hat{y}_{\text{LRT}}, \hat{\alpha}, \hat{\beta}^+, \hat{p}^+ \right) \quad (1)$$

2 Input Representation & Dual Tokenization

2.1 Dual Codon-AA Tokenization

Each alignment entry (i, j) at species i and codon site j is tokenized across two parallel feature channels:

- **Codon Vocabulary** ($|\mathcal{C}| = 64$): Sense codons $0 \dots 63$. Stop codons, gaps, and unknown nucleotides are assigned index 64.
- **Amino Acid Vocabulary** ($|\mathcal{A}| = 20$): Standard amino acids $0 \dots 19$. Unknowns / gaps are assigned index 20.

2.2 Embedding Layer

$$\mathbf{e}_{\text{codon}} = \mathbf{E}_{\text{codon}}[c_{i,j}] \in \mathbb{R}^{64} \quad (2)$$

$$\mathbf{e}_{\text{aa}} = \mathbf{E}_{\text{aa}}[a_{i,j}] \in \mathbb{R}^{64} \quad (3)$$

$$\mathbf{x}_{i,j}^0 = \text{Concat}(\mathbf{e}_{\text{codon}}, \mathbf{e}_{\text{aa}}) + \mathbf{P}_{\text{pos}}[j] + \mathbf{W}_{\text{mds}} \mathbf{m}_i \in \mathbb{R}^{128} \quad (4)$$

where $\mathbf{P}_{\text{pos}} \in \mathbb{R}^{L \times 128}$ is a 1D positional embedding and $\mathbf{m}_i \in \mathbb{R}^4$ represents Multidimensional Scaling (MDS) coordinates derived from the phylogenetic tree distance matrix.

3 Axial Transformer Backbone

Axomeme 2.0 incorporates 4 interleaved axial attention blocks to achieve $O(N \cdot L)$ scaling:

3.1 Column-Attention Layer (Codon Spatial Context)

Captures local sequence context along the codon dimension L :

$$\mathbf{H}_{\text{col}} = \text{MultiHeadAttention}(\mathbf{X}, \mathbf{X}, \mathbf{X}) \in \mathbb{R}^{(B \cdot N) \times L \times 128} \quad (5)$$

3.2 Phylo-Row-Attention Layer (Taxa Context)

Captures phylogenetic variation across species N with distance normalization:

$$\mathbf{D}_{\text{norm}} = \frac{\mathbf{D}}{\bar{D} + 10^{-4}} \quad (6)$$

Pairwise synonymous ($\mathbf{M}_{\text{syn}}$) and non-synonymous ($\mathbf{M}_{\text{nonsyn}}$) transition masks derived from the genetic code are injected directly into the attention logits.

4 Multi-Stream Species Representation Pooling

To aggregate representations across N species into a single site vector $\mathbf{h}_j \in \mathbb{R}^{128}$:

$$\mathbf{h}_{\text{mean}} = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_{i,j}, \quad \mathbf{h}_{\text{max}} = \max_{i=1}^N (\mathbf{x}_{i,j}) \quad (7)$$

$$\mathbf{h}_{\text{attn}} = \sum_{i=1}^N \text{Softmax}(\mathbf{w}_a^\top \mathbf{x}_{i,j}) \mathbf{x}_{i,j}, \quad \mathbf{h}_{\text{diff}} = \text{ReLU}(\mathbf{h}_{\text{max}} - \mathbf{h}_{\text{mean}}) \quad (8)$$

$$\mathbf{h}_j = \text{Linear}(\text{Concat}(\mathbf{h}_{\text{mean}}, \mathbf{h}_{\text{max}}, \mathbf{h}_{\text{attn}}, \mathbf{h}_{\text{diff}})) \quad (9)$$

5 Rank-Consistent CORAL Head & Expected Value Decoder

To guarantee strict rank consistency ($b_0 > b_1 > \dots > b_{11}$):

$$b_k = b_0 - \sum_{j=1}^k \text{Softplus}(\theta_j), \quad z_k = \mathbf{w}^\top \mathbf{h}_j + b_k \quad (10)$$

where $p_k = \sigma(z_k) = P(\text{LRT} > E_{k+1})$.

Continuous predictions are decoded via the **Smooth Expected Value Decoder**:

$$P(\text{Bin } 0) = 1.0 - p_0, \quad P(\text{Bin } m) = p_{m-1} - p_m, \quad P(\text{Bin } 12) = p_{11} \quad (11)$$

$$\hat{y}_{\text{LRT}} = \mathbb{E}[\text{LRT}] = \sum_{m=0}^{12} P(\text{Bin } m) \cdot \mu_m \quad (12)$$

6 Performance Summary

Model / Benchmark	Mean AUC-ROC	Mean AUC-PR	False Positive Rate
HyPhy MEME (Likelihood)	0.8100	0.3500	5.0%
PAML CODEML (Likelihood)	0.7800	0.3000	6.2%
Axomeme 2.0 (Epoch 8 Rebalanced)	0.7772	0.2197	0.46%

Table 1: Overall benchmark metrics for Axomeme 2.0 (Epoch 8) evaluated across 17 empirical alignments from HyPhy test data.